

Introduction to Retrieval-Augmented Generation

What is RAG?

Retrieval-Augmented Generation, commonly called RAG, is an architecture that combines information retrieval with a large language model. Instead of asking a language model to answer only from its learned parameters, a RAG application retrieves relevant information from an external knowledge base and provides that information as context.

Basic RAG Pipeline

A typical RAG pipeline contains document loading, text splitting, embedding generation, vector storage, retrieval, prompt construction, and answer generation. Documents are divided into smaller chunks and converted into vector embeddings. When a user asks a question, the system searches for chunks that are semantically similar to the query.

Benefits

RAG can help applications answer questions using private or frequently changing information. It can also provide source context to the model and reduce the need to retrain a language model whenever the knowledge base changes.

Example

A company can build a RAG assistant over internal documentation. An employee could ask about a company policy, and the system could retrieve the relevant policy sections before asking the language model to generate a natural-language answer.